Synopsis

Disease Risk Prediction Using Machine Learning Techniques

Project ID: **PY-GT-48**

Problem ID: **1**

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Repo→https://github.com/SnehilJoshi007/PBL-AIML

Problem Description

Machine learning is widely used in healthcare to analyze patient data and assist in early disease prediction. Many diseases such as heart disease can be predicted by studying clinical parameters like age, blood pressure, cholesterol level, blood sugar, heart rate, and other medical attributes. Manual analysis of large patient records is difficult and time-consuming, so an automated system can be very useful.

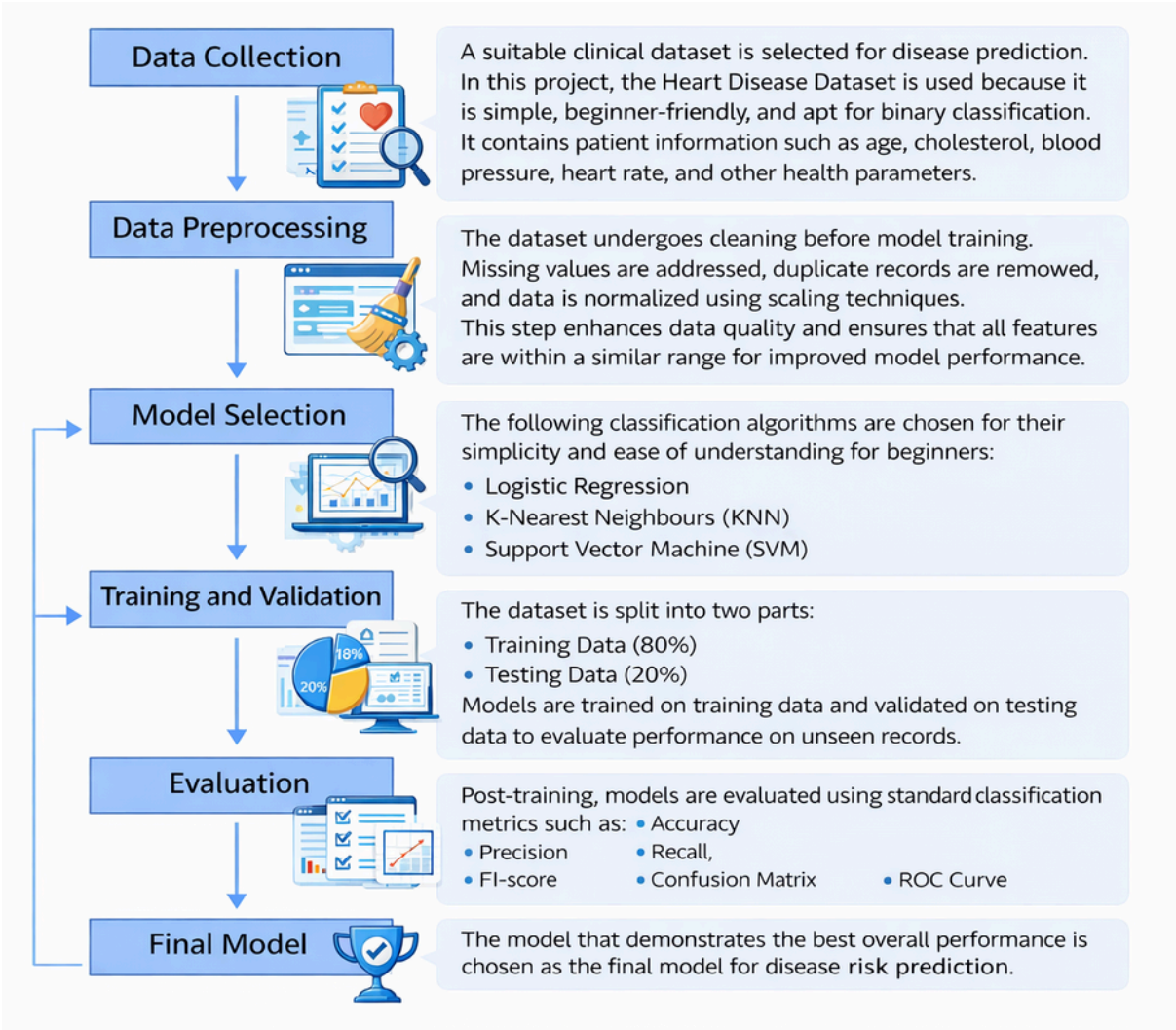
The objective of this project is to build a simple machine learning classification system that predicts whether a patient is at risk of disease or not using clinical data. The project uses a healthcare dataset and applies preprocessing techniques such as data cleaning and normalization to improve data quality. Three machine learning algorithms—Logistic Regression, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM)—are used for training and testing.

The models are evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. This project demonstrates a complete beginner-level machine learning pipeline and shows how AI can support decision-making in the healthcare field.

Steps of Implementation

This project adheres to a straightforward machine learning pipeline. Each phase of implementation is outlined below:

1. **Data Collection**
   A suitable clinical dataset is selected for disease prediction. In this project, the Heart Disease Dataset is used because it is simple, beginner-friendly, and apt for binary classification. It contains patient information such as age, cholesterol, blood pressure, heart rate, and other health parameters.
2. **Data Preprocessing**
   The dataset undergoes cleaning before model training. Missing values are addressed, duplicate records are removed, and data is normalized using scaling techniques. This step enhances data quality and ensures that all features are within a similar range for improved model performance.
3. **Model Selection**
   The following classification algorithms are chosen for their simplicity and ease of understanding for beginners:
   - Logistic Regression
   - K-Nearest Neighbours (KNN)
   - Support Vector Machine (SVM)
4. **Training and Validation**
   The dataset is split into two parts:
   - Training Data (80%)
   - Testing Data (20%)
5. Models are trained on training data and validated on testing data to evaluate performance on unseen records.
6. **Evaluation**
   Post-training, models are evaluated using standard classification metrics such as:
   - Accuracy
   - Precision
   - Recall
   - F1-score
   - Confusion Matrix
   - ROC Curve
7. **Final Model**
   The model that demonstrates the best overall performance is chosen as the final model for disease risk prediction.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18%
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• Precision • Recall,
• FI-score • Confusion Matrix • ROC Curve
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Expected Modules / Libraries to be Used

The following Python libraries and key Scikit-learn modules are anticipated for use in this project:

1. **NumPy**
   Utilized for numerical operations and efficient handling of arrays in Python.
2. **Pandas**
   Used for loading the dataset from CSV files and for organizing, cleaning, and manipulating tabular data.
3. **Matplotlib**
   Employed for plotting graphs such as ROC curves and performance visualizations.
4. **Seaborn**
   Primarily used for advanced data visualization, such as drawing the confusion matrix heatmap.
5. **Scikit-learn (sklearn)**
   The core machine learning library used for preprocessing, model training, testing, and evaluation.

Important Modules from Scikit-learn

- **train_test_split**: Divides the dataset into training and testing data.
- **StandardScaler**: Normalizes or scales input features to ensure they are within a similar range.
- **LogisticRegression**: Implements the Logistic Regression classifier.
- **KNeighborsClassifier**: Implements the K-Nearest Neighbours (KNN) classifier.
- **SVC**: Implements the Support Vector Machine (SVM) classifier.
- **accuracy_score**: Calculates model accuracy.
- **precision_score**: Computes model precision.
- **recall_score**: Determines model recall.
- **f1_score**: Calculates the model's F1-score.
- **confusion_matrix**: Generates the confusion matrix.
- **roc_curve**: Creates the ROC curve for model evaluation.

Expected Evaluation Metrics and Plots to be Used

The following evaluation metrics and plots will be employed to compare machine learning model performance:

1. **Accuracy**
   Measures the correctness of predictions made by the model out of the total predictions.
2. **Precision**
   Assesses how many predicted positive cases are actually correct, crucial for reducing false positives.
3. **Recall**
   Determines the number of actual positive cases correctly identified by the model, which is vital in medical contexts where missing a disease case can be risky.
4. **F1-Score**
   Provides a balanced measure of model performance as the harmonic mean of precision and recall.
5. **Confusion Matrix**
   Displays the number of correct and incorrect predictions, aiding in understanding true positives, true negatives, false positives, and false negatives.
6. **ROC Curve**
   Used to compare model classification performance at different thresholds.
7. **AUC Score (Optional but Recommended)**
   Indicates how well the model can separate classes; a higher AUC suggests better performance.

Plots to be Used

- Confusion Matrix Heatmap
- ROC Curve Plot
- Model Comparison Bar Graph (Accuracy / F1-score)

Required Topics from the Subject

1. **Strings**
   Used for managing text data, handling column names, labels, file paths, and displaying output messages.
2. **Pointers**
   Though primarily a concept in C programming, understanding pointers aids in efficient data handling and memory management.
3. **Files**
   Essential for reading and writing data, as the clinical dataset is stored in a CSV file and loaded into Python for processing.
4. **Arrays**
   Used to store multiple values of the same type, crucial for storing numerical feature values and model input data.
5. **Functions**
   Help divide the program into smaller, reusable blocks. Useful for preprocessing, training models, and evaluating performance.
6. **Conditional Statements**
   Facilitate decision-making in a program, such as checking for missing values or selecting the best model based on accuracy.

Platform Required

The project can be implemented using the following platform:

- **Jupyter Notebook**: Ideal for writing and executing Python code incrementally. It is user-friendly and well-suited for machine learning experiments.

Books and Link Sources

- Scikit-learn Documentation – https://scikit-learn.org
- Pandas Documentation – https://pandas.pydata.org
- NumPy Documentation – https://numpy.org
- Matplotlib Documentation – https://matplotlib.org

Dataset-

Repo-*https://github.com/SnehilJoshi007/PBL-AIML.*

📊 Dataset Overview

Parameter	Details
Purpose	Synthetic dataset for disease risk prediction using patient clinical data. Designed for data cleaning, preprocessing, feature selection, and machine learning classification.
Total Rows	1800
Total Columns	12
Data Quality Design	Includes missing values (~6%) and some noise/outliers to simulate real-world clinical data inconsistencies.

📁 Column Details

Column	Description	Typical Unit	Missing Values (~)
Age	Patient age	Years	~108
Gender	0 = Female, 1 = Male	Binary	~108
BMI	Body Mass Index	kg/m^2	~108
Blood_Pressure	Systolic blood pressure	mmHg	~108
Cholesterol	Serum cholesterol level	mg/dL	~108
Glucose	Blood sugar level	mg/dL	~108
Heart_Rate	Heart beats per minute	bpm	~108
Smoking	Smoking status	Binary	~108
Alcohol_Intake	Weekly alcohol consumption	Units/week	~108
Physical_Activity	Weekly exercise duration	Hours/week	~108
Family_History	Genetic predisposition	Binary	~108
Disease_Risk	Target variable (0 = Low, 1 = High)	Binary	~108

⚡ Quick Summary

Column	Min	Max	Comment
Age	18	79	Wide age distribution across adults
BMI	~15	~40+	Includes underweight to obese cases
Blood Pressure	90	180+	Contains hypertension cases
Cholesterol	150	300+	Some high-risk values present
Glucose	70	190	Includes diabetic range
Heart Rate	55	115	Includes abnormal heart rates
Physical Activity	0	10	Covers sedentary to active lifestyles
Disease Risk	0	1	Balanced classification target

1	Age (years)	Gender	BMI (kg/m^2)	Blood_Pressure (mmHg)	Cholesterol (mg/dL)	Glucose (mg/dL)	Heart_Rate (bpm)	Smoking	Alcohol_Intake (units/week)	Physical_Activity (hours/week)	Family_History	Disease_Risk
2	34	male	27.9	94/119	277	104	111		4	5.8	no	Low
3	72	Female	25.7	163/109	264	122	81		4	6.4	Yes	High
4	29	Others		161/108	248	103	90	no	1	3.3	unknown	High
5	18	female	27.9	115/112	191	127	64	N	6	9.1	N	
6	34	male	25.5	169/117	152		79	Yes	6	8.4	N	Medium
7	19	Others	19	106-68	160	163	56	No	1	6.7	Yes	High
8	27	F	33.8	139/83	174	170	113	N	4	0	yes	Medium
9	18	Others	36.3	94-63			58		8	5.5	N	Low
10	68	Female	32.4	164/99	258	175	71	No	7	1.3	N	High
11	39	Others	18.2	166/62		149	90	N	4	2.7	Yes	High
12	30		30.8	127/101	246	144	78	no	3	9.8	Y	Medium
13	32		20.5	159-75	244	124	118	Y	0	3.5	Y	Low
14	20	Others	21.3	134-81	236	107	112		9	8.3	Y	Low
15	18	M	32.6	127/92	193	137	96	No	11	5.3	no	Low
16	22	male	29.5	107/72	248	89	105	yes	11	9.5	yes	Medium
17	22	Others	24.6	100/101	222	191	91	No	6	8.9	No	Low
18	35		29.8	151/119	212	73	112	Y	2	9.7	No	Low
19	38	M	33.8	106/118	180	141	70	Yes	2	2.5	yes	Low
20	19	Male	26.8	147/112	228	170		Yes	9	9.3	yes	Low
21	75	Others	36.7	162/116	245	88	113	N	5	1	Yes	Medium
22	20	male	16.5	151/83	254	86	113	Y	5	1.4	Y	Medium
23	22	Male	30.3	168/119	311	160	97	yes	4	3	no	High
24	22	Female	17.1	139/66	174	156	55	No	10	8.3	no	High
25	21	M		129/61	273	190	64	no	4	2.7	yes	High
26	68	M	25.1	96-76	244	122	118	no	9	2.1	Y	
27	41	F	23.5	109/119	172	181	88	N	5	1.7	Yes	Medium
28		male	28.1	143/91	293	193	68	Yes	2	3.5		Medium
29	45	Others	26.3	154/84	223	129	79	Yes	2	0.7	no	High
30	21	Others	25.7	175/86	243	83	81	Yes	7	5.2	Y	Low